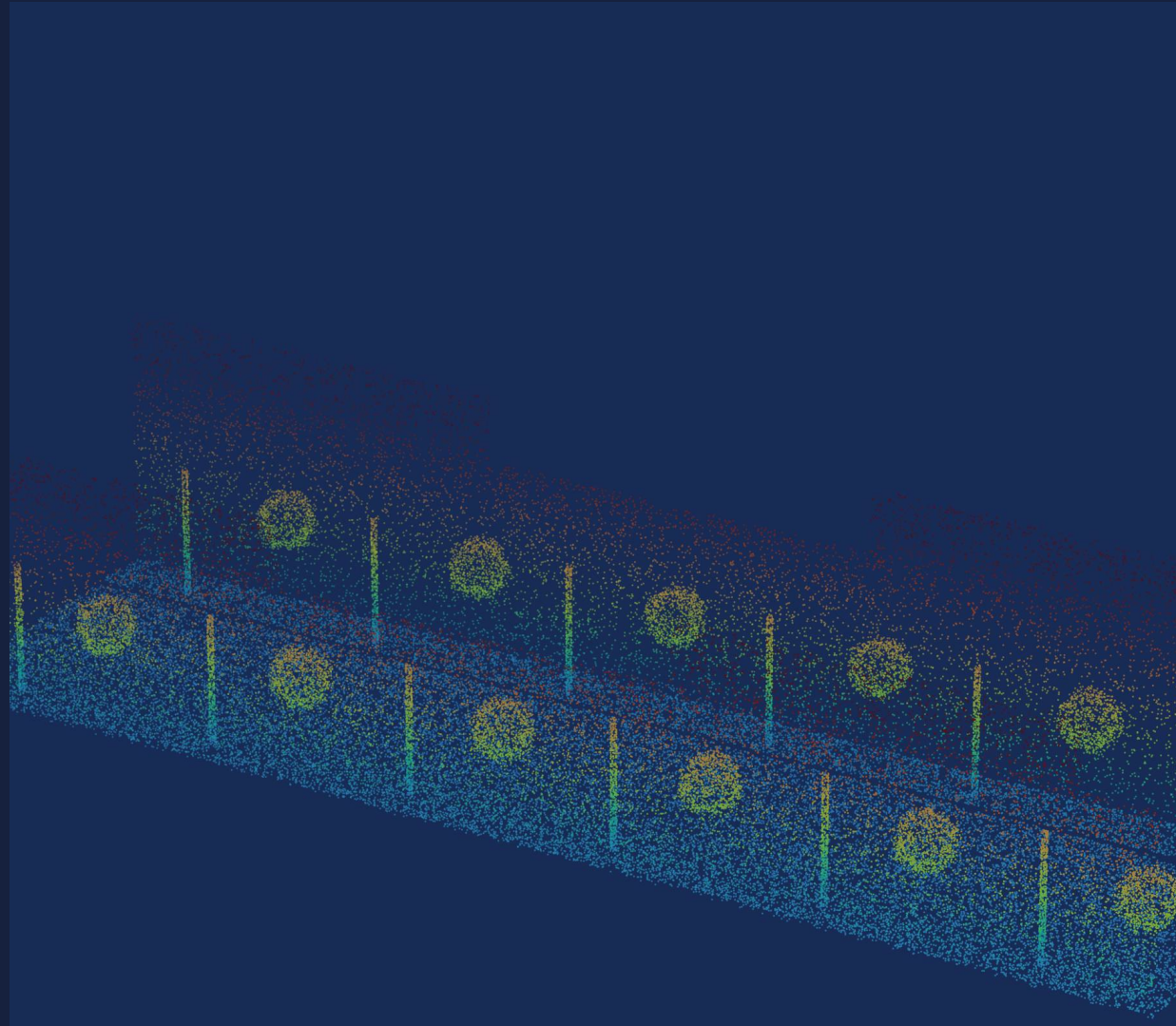
SEMINAR 2 • ADVANCED

Engineering, Error Budgets and Standards in Numbers

Mobile laser scanning for surveyors: the equations, the budgets and the tests behind a survey-grade deliverable.

D.S. Banks, PSM



LEARNING OBJECTIVES

By the end of today you will be able to. . .

- Compute a range, its timing sensitivity and its ambiguity interval
- Apply rotations and the direct georeferencing equation to place a point
- Predict IMU drift through a GNSS outage from datasheet biases
- Work a Kalman update, an ICP fit and a small factor graph by hand
- Propagate errors with the variance law and check them by Monte Carlo
- Build a complete error budget and find the term that limits it
- Convert FDOT 95% tolerances to ASPRS RMSE and run one test for both
- Design control and checkpoint layouts from the error budget

TODAY

Agenda

Time	Session	Reference Book
0:00–0:15	Welcome, recap, the road map	Module 9
0:15–1:05	1. The single measurement in numbers	Modules 3, 4, 5
1:05–1:55	2. Pose, IMU and calibration math	Modules 6, 7, 8, 9
1:55–2:10	Break	
2:10–3:20	3. Estimation: how the trajectory is computed (core block)	Modules 10, 11, 12
3:20–3:50	Lunch	
3:50–4:50	4. Error propagation and error budgets	Module 14
4:50–5:35	5. Hardware in numbers	Modules 15, 16, 17
5:35–5:45	Break	
5:45–6:35	6. The standards in numbers	Modules 13, 19, 21
6:35–7:05	Problem set and evaluation	

HOW TODAY WORKS

Seminar 1 gave the concepts. Today gives the numbers.

	Seminar 1	Seminar 2
Level	concepts, no calculations	calculations on every topic
Text	Seminar edition (R1)	Reference Book (R0.1)
Book material	Simple Approach, Big Idea, Field Tip boxes	Engineer’s View, Key Equation, Worked Example boxes
Tools	none	calculator and the workbook
Assessment	50 conceptual questions	20 worked problems

Workbook: one sheet per module, M02 to M21, plus the Monte Carlo sheet

One equation runs the whole day

$$\mathbf{r}_W = \mathbf{p}_W(t) + \mathbf{R}_B^W(t)\left(\mathbf{R}_S^B\,\mathbf{r}_S + \mathbf{l}^B\right)$$

$\mathbf{r}_S$	the laser point in the sensor frame: range and two angles (Session 1)
$\mathbf{R}_S^B,\ \mathbf{l}^B$	boresight and lever arm: calibration (Session 2)
$\mathbf{R}_B^W(t)$	attitude from the IMU (Sessions 2 and 3)
$\mathbf{p}_W(t)$	position from GNSS, IMU and SLAM (Session 3)
$\mathbf{r}_W$	the map coordinate, with its error budget (Sessions 4 to 6)

- Every term carries an error
- Fifteen uncertain inputs in all
- The budget in Session 4 adds them up

SESSION 1 • 0:15–1:05 • MODULES 3, 4, 5

The single measurement in numbers

Time of flight and clock precision, range ambiguity, the beam footprint, and turning a range and two angles into X, Y, Z.

MODULE 3 • KEY EQUATION

Time-of-flight range

$$\rho = \frac{c\,\Delta t}{2}, \qquad \delta\rho = \frac{c\,\delta t}{2} \approx 0.15\ \text{m per ns}$$

Given	an echo returns after 400 ns
Range	$\rho \approx 59.94$ m
Sensitivity	1 ns of timing error $\rightarrow$ 15 cm of range
So	millimeter ranging needs picosecond timing

15 cm

of range for every nanosecond of round-trip time

- Divide by two: the light goes out and back
- Air slows light by about 300 ppm

MODULE 3 • KEY EQUATION

How fast can we fire? Maximum unambiguous range

$$\rho_{\max} = \frac{c}{2\, f_{\mathrm{PRF}}}$$

1 MHz	$2.997 \times 10^8 / (2 \times 10^6) = 149.9$ m
2 MHz	74.9 m
2.4 MHz	62.4 m: a 500 ft (152.4 m) echo is in zone 3

149.9 m

safe range at one million pulses per second

- Faster firing gives denser points and a shorter safe range
- A wrong zone moves a point by a whole multiple of the interval: a blunder, not noise

Book: Modules 3 and 17 • Workbook: sheets M03 and M17

MODULE 4 • KEY EQUATION

A point is really a spot: beam footprint

$$D(\rho) \approx d_0 + \rho\,\gamma$$

Given	exit diameter $d_0 = 5$ mm, divergence $\gamma = 0.3$ mrad
At 50 m	$0.005 + 50 \times 0.0003 = 2.0$ cm
At 100 m	$0.005 + 100 \times 0.0003 = 3.5$ cm
Wide beam	0.35° divergence: 12.7 cm at 20 m

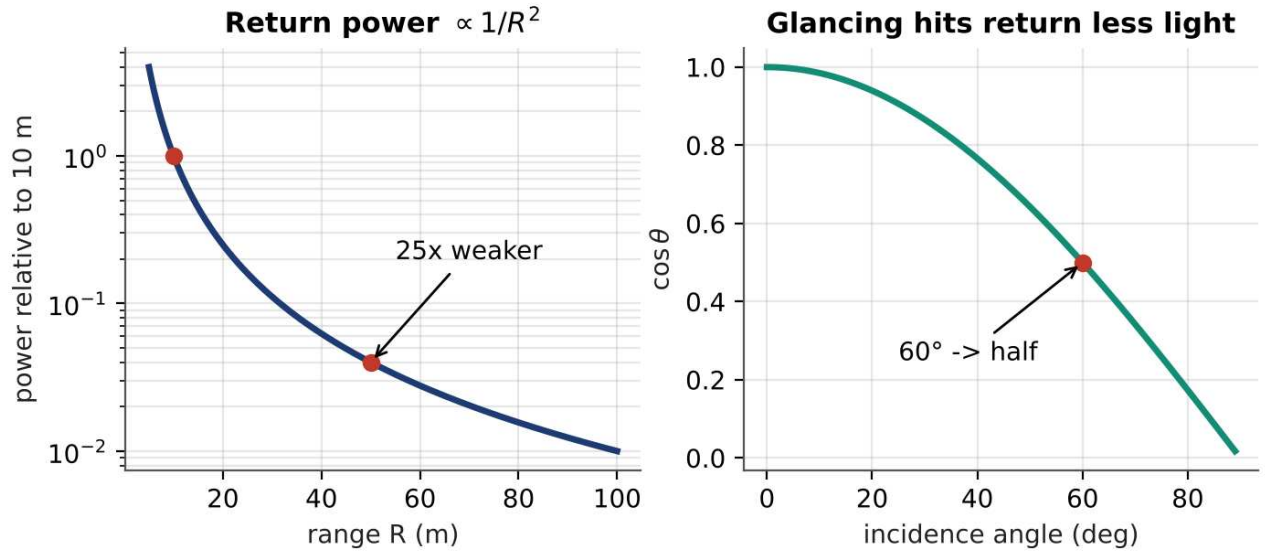
3.5 cm

spot diameter at 100 m for a survey-grade beam

- An edge cannot be located more sharply than about half the spot
- Horizontal accuracy is limited by the footprint, not only by the trajectory

Book: Modules 4 and 15 • Workbook: sheet M04 Beam

Echo strength sets the usable range



25 m

usable on fresh asphalt for a sensor rated 35 m on a 10% target

- Received power falls as $1/R^2$ and with $\cos\theta_i$
- Range scales with $\sqrt{\text{reflectivity}}$: $35 \times \sqrt{5/10} = 25$ m
- Plan the swath on the darkest surface you must map

MODULE 5 • KEY EQUATION

A range and two angles become X, Y, Z

$$x_s = \rho \cos V \cos H, \quad y_s = \rho \cos V \sin H, \quad z_s = \rho \sin V$$

Given	$\rho = 25$ m, $H = 40°$, $V = 15°$
Level part	$\rho \cos V = 24.15$ m
Result	$(x_s, y_s, z_s) = (18.50,\ 15.52,\ 6.47)$ m
Check	$\sqrt{18.50^2 + 15.52^2 + 6.47^2} = 25.00$ m

25.00 m

the length check: a conversion never changes the range

- The same geometry as a total-station shot
- The inverse uses atan2, which keeps the quadrant right

MODULE 5 • KEY EQUATION

Point spacing and point rate

$$\Delta_{\text{along}} = \frac{v}{f_{\text{rot}}}, \qquad \Delta_{\text{across}} \approx \rho\,\Delta\theta, \qquad N = N_{\text{ch}}\,N_{\text{az}}\,f_{\text{rot}}$$

Given	16 channels, 0.2° step, 10 Hz, 15 m/s, target at 20 m
Along track	$15/10 = 1.5$ m between sweeps
Across track	$20 \times 0.003491 = 0.070$ m
Point rate	$16 \times 1800 \times 10 = 288\,000$ points/s
Slow to 5 m/s	0.5 m between sweeps

1.5 m

between scan lines at 15 m/s and 10 Hz

- Fine for pavement, too coarse for curb lines
- Slow down, tilt the head, or add channels

DENSITY HAS A TARGET

20 pts/ft²

FDOT Type A minimum, about 215 points/m^2

10 pts/ft²

FDOT Type B minimum, about 108 points/m^2

0.22 ft

point spacing at 20 points/ft^2: 2.7 inches

Multiply points/ft^2 by 10.764 to get points/m^2. Check density on the delivered data, at the farthest range used for design.

SESSION 2 • 1:05–1:55 • MODULES 6, 7, 8, 9

Pose, IMU and calibration math

Rotation matrices and the small-angle rule, IMU drift and the tilt error, timing and de-skewing, lever arm and boresight.

MODULE 6 • KEY EQUATION

Rotating a point

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \qquad \mathbf{R} = \mathbf{R}_z(\psi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_x(\phi)$$

Given	rotate $(3,\ 4)$ by 30° counter-clockwise
Result	$(0.598,\ 4.964)$
Check	length is still 5: a rotation never changes lengths
In 3-D	yaw, then pitch, then roll (ZYX order). Order matters

$$\psi = 90^\circ - \alpha$$

yaw is not azimuth: convert before using a heading

- Rigid-body transform: rotate, then shift
- Confirm the frame convention with the manufacturer

Why tiny angle errors matter



7.0 cm

from a 0.05° heading error at 80 m

- Sideways error $= \rho \times \delta\alpha$ (in radians)
- 0.01° at 50 m: 8.7 mm
- 0.1° at 100 m: 17.5 cm

MODULE 7 • KEY EQUATION

Why IMUs drift: biases integrate

$$\delta x(t) = \tfrac{1}{2}\, b_a\, t^2, \qquad \delta y(t) \approx \tfrac{1}{2}\, v\, b_g\, t^2$$

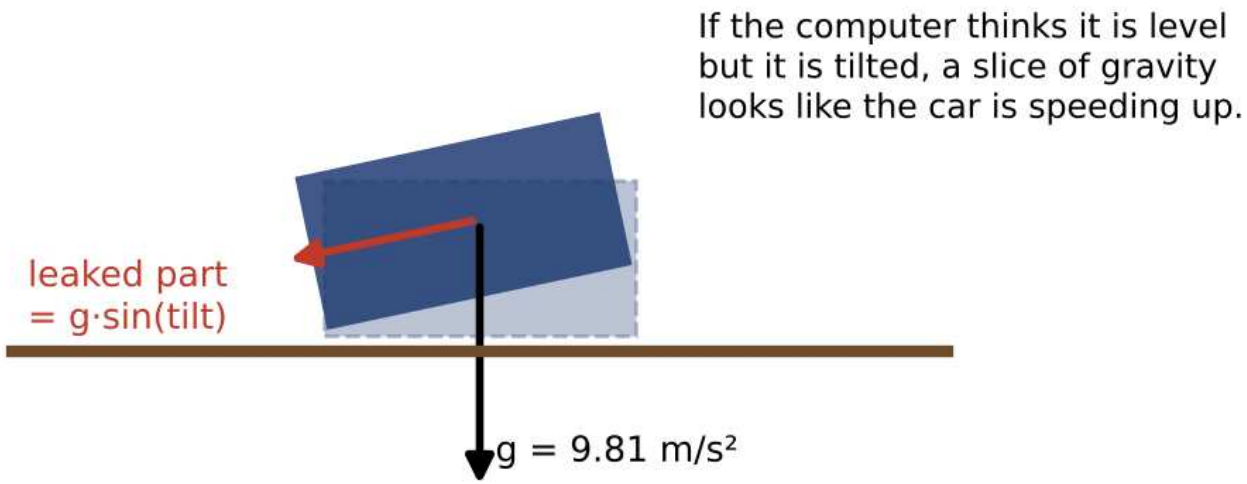
Given	gyro bias 1°/h, accelerometer bias 0.5 mg, 15 m/s, 60 s unaided
Gyro	heading error 0.0167°; sideways 0.131 m
Accelerometer	$\tfrac{1}{2} \times 0.004903 \times 60^2 = 8.83$ m along track
After 30 s	2.2 m along track

8.83 m

along-track drift in one minute, tactical-grade IMU, no aiding

- Time squared: doubling the outage quadruples the drift
- No survey system coasts on its IMU for long

A tilt error leaks gravity into acceleration



0.86 m

after 10 s from a 0.1° tilt error

- Leaked acceleration $= g \sin 0.1^\circ = 0.0171$ m/s^2
- That is 1.7 mg: worse than the accelerometer bias
- Attitude must be known very well, or gravity becomes the biggest error

MODULE 8 • KEY EQUATION

What a timing error costs

$$\delta_{\text{along}} = v\,\delta t \qquad\qquad \delta_{\text{lat}} \approx \rho\,\dot{\psi}\,\delta t$$

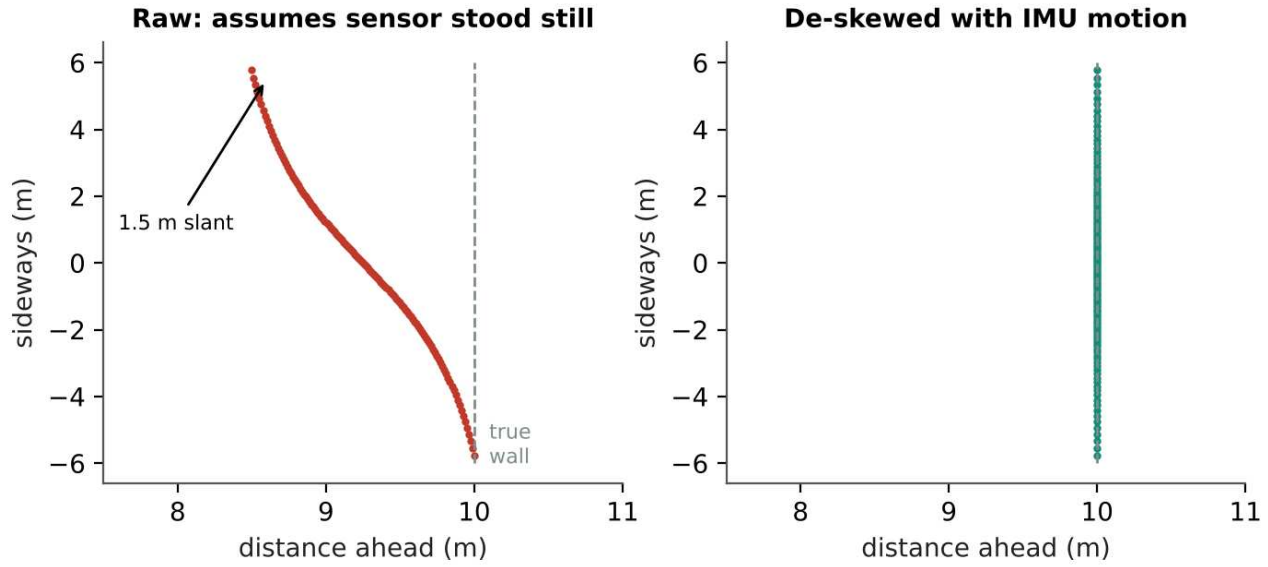
Given	$\delta t = 1$ ms; 20 m/s; turning at 30°/s; point 50 m away
Along track	$20 \times 0.001 = 2.0$ cm
Rotation	$30 \times 0.001 = 0.03° = 0.000524$ rad
Sideways	$50 \times 0.000524 = 2.6$ cm

2.6 cm

sideways in a turn from a 1 ms clock offset

- Corners look smeared but straights look fine: the timing-error signature
- Synchronize with PPS or PTP and verify in a turn

De-skewing: every point gets its own pose

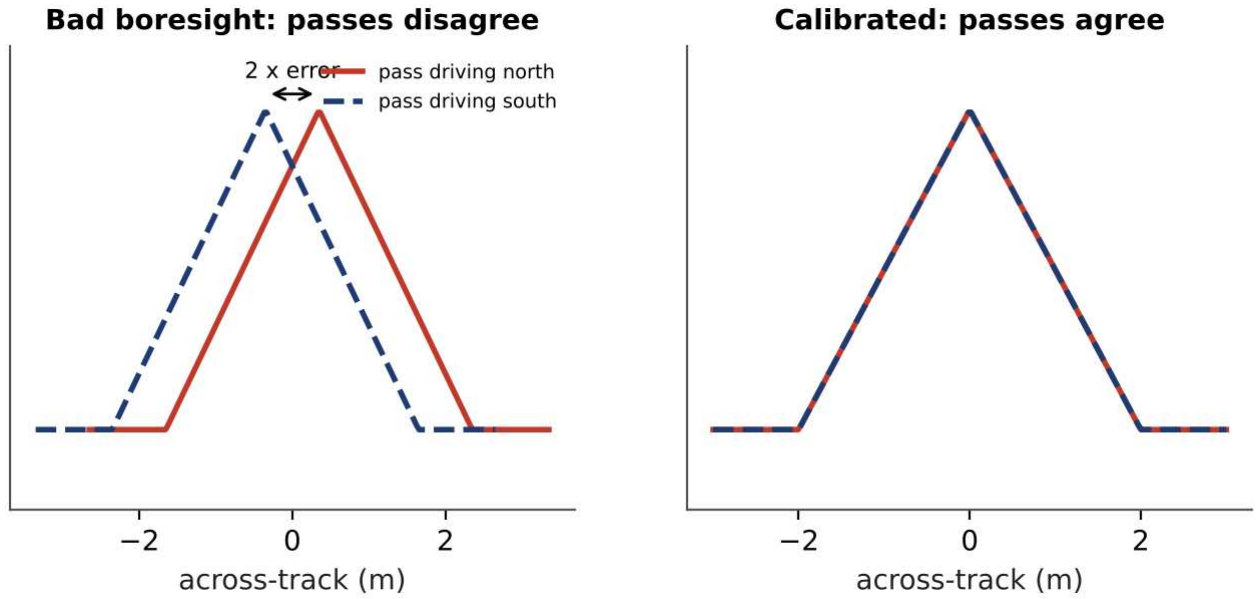


1.5 m

between the first and last point of one 10 Hz sweep at 15 m/s

- A point fired 0.075 s into the sweep moved $15 \times 0.075 = 1.125$ m
- Each point is moved back using its own timestamp
- Needs a per-point time field and the IMU

Boresight from two passes



0.040°

from a 4.2 cm offset at 30 m between opposite passes

- Opposite passes double the effect: $\delta\psi = \Delta/(2\rho)$
- $0.042/60 = 7.0 \times 10^{-4}$ rad
- Lever arm error: fixed size. Boresight error: grows with range

THREE MULTIPLICATIONS TO REMEMBER

7.0 cm	2.6 cm	0.86 m
0.05° of heading or boresight at 80 m	1 ms of timing at 50 m in a 30°/s turn	0.1° of tilt after 10 s unaided

Angle error times range. Timing error times speed, or times turn rate times range. Tilt error times gravity times time squared.

SESSION 3 • 2:10–3:20 • MODULES 10, 11, 12

Estimation: how the trajectory is computed

The Kalman filter, scan matching, and the factor graph inside LIO-SAM. Each one is worked once by hand.

MODULE 10 • KEY EQUATION

The Kalman update in one dimension

$$K = \frac{\bar{\sigma}^2}{\bar{\sigma}^2 + \sigma_z^2}, \qquad \mu = \bar{\mu} + K(z - \bar{\mu}), \qquad \sigma^2 = (1 - K)\,\bar{\sigma}^2$$

Prediction	$\bar{\mu} = 100.0$ m, $\bar{\sigma} = 0.50$ m
Measurement	$z = 100.4$ m, $\sigma_z = 0.20$ m
Gain	$K = 0.25/(0.25 + 0.04) = 0.8621$
Update	$\mu = 100.0 + 0.8621 \times 0.4 = 100.345$ m
New sigma	$\sqrt{(1 - 0.8621) \times 0.25} = 0.186$ m

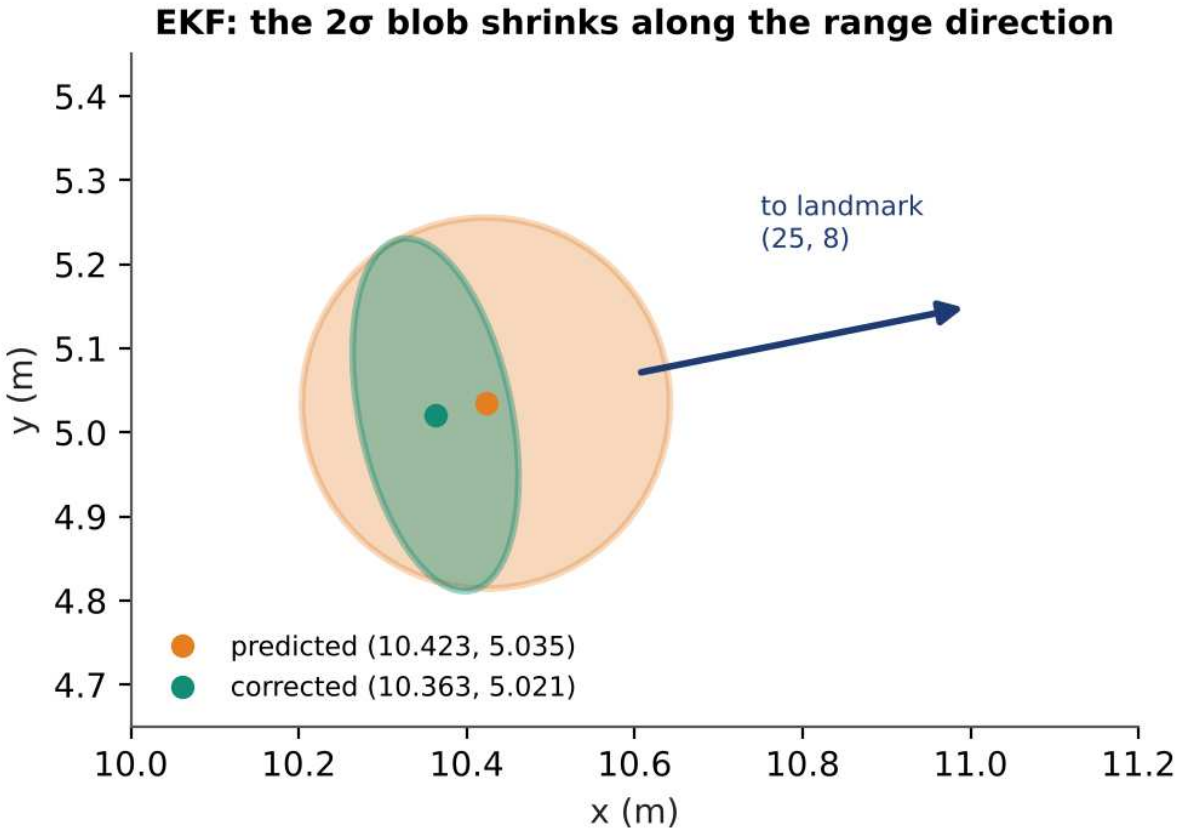
0.186 m

better than the measurement’s own 0.20 m

- The answer lands between the two, closer to the more certain one
- Fusion beats the best single source

Book: Module 10 • Workbook: sheet M10 SLAM

Moving grows uncertainty; a landmark shrinks it



11.0 to 4.9 cm

in x, after one range and bearing to a pole

- Predict: $\bar{\mathbf{P}} = \mathbf{G}\,\mathbf{P}\,\mathbf{G}^{\mathsf{T}} + \mathbf{Q}$
- y barely improved: the pole is almost straight ahead
- A range only constrains the direction it points

MODULE 11 • KEY EQUATION

Scan matching: Iterative Closest Point

$$\min_{\mathbf{R},\mathbf{t}} E = \sum_{i=1}^{N} \left\| \mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_{c(i)} \right\|^2$$

1. Pair	each point with its nearest neighbor in the other scan
2. Fit	centroids, then the best rotation and shift in closed form
3. Move	apply the rotation and shift
4. Repeat	until the pairs stop changing
Worked	four matched points recover a 10° rotation and a (1.0, 0.5) m shift

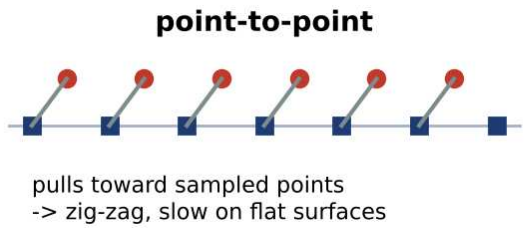
10°

and (1.0, 0.5) m: exactly how the test scan was made

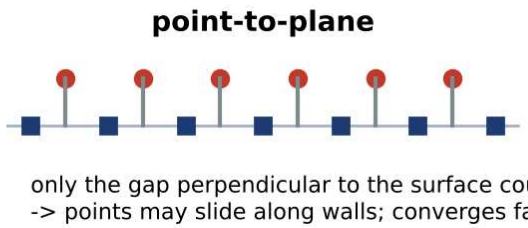
- Needs a good starting guess: the IMU supplies it
- Lab 3 runs this in 30 lines of Python

Book: Module 11 and Lab 3 • Workbook: sheet M11 Scan Matching

Point-to-plane lets points slide along walls



pulls toward sampled points
-> zig-zag, slow on flat surfaces

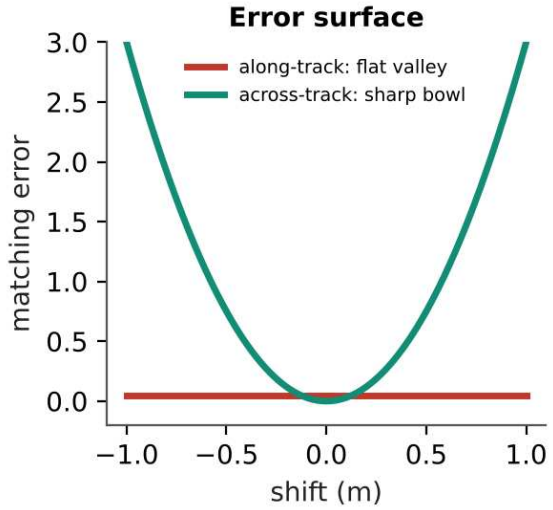


only the gap perpendicular to the surface counts
-> points may slide along walls; converges faster

- Point-to-point pulls toward sampled points: slow on flat surfaces
- Point-to-plane counts only the gap normal to the surface
- LOAM and LIO-SAM match edge and planar features only
- Minimize $\sum d_e + \sum d_p$ by Gauss–Newton

The hallway effect: nothing to grip along the corridor





- The matching error has a flat valley along the corridor
- Any forward shift fits equally well
- Add control or targets, cross-ties, or temporary objects
- Check along-track distances against a taped measurement

MODULE 12 • KEY EQUATION

A factor graph is a least-squares adjustment

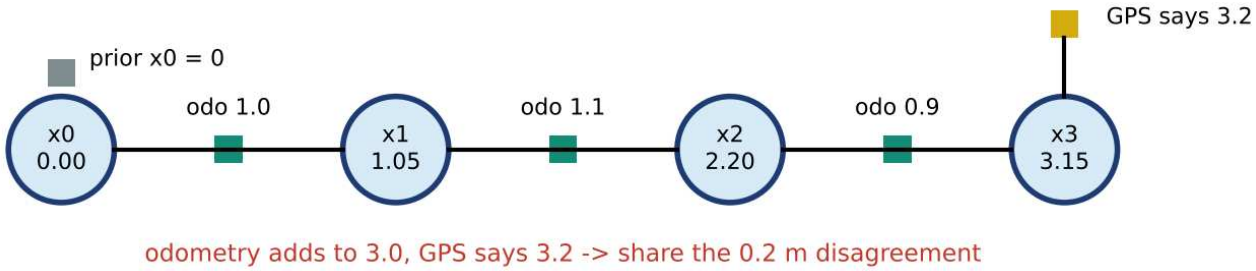
$$\mathbf{X}^* = \arg\max_{\mathbf{X}} \prod_k \phi_k(\mathbf{X}_k) = \arg\min_{\mathbf{X}} \sum_k \left\| \mathbf{h}_k(\mathbf{X}_k) - \mathbf{z}_k \right\|^2_{\mathbf{\Sigma}_k}$$

Unknowns	the poses: circles in the graph
Factors	the measurements that tie them together: GNSS, IMU, scan matching, loop closures
Weights	each factor's covariance $\mathbf{\Sigma}_k$
Solution	the poses that best fit every measurement at once

- Gaussian noise turns the most probable answer into weighted least squares
- The same mathematics as a network adjustment
- A late loop closure corrects early poses

MODULE 12 • WORKED BY HAND

Four poses, three odometry legs, one GPS fix



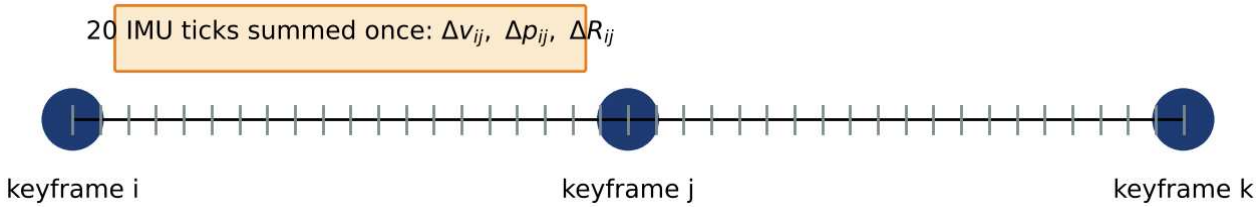
0.20 m

misclosure, shared equally by four equally weighted measurements

- Odometry: 1.0, 1.1, 0.9 m. GPS says 3.2 m
- Solution: $x_1 = 1.05$, $x_2 = 2.20$, $x_3 = 3.15$ m
- A traverse adjustment, done by a factor graph

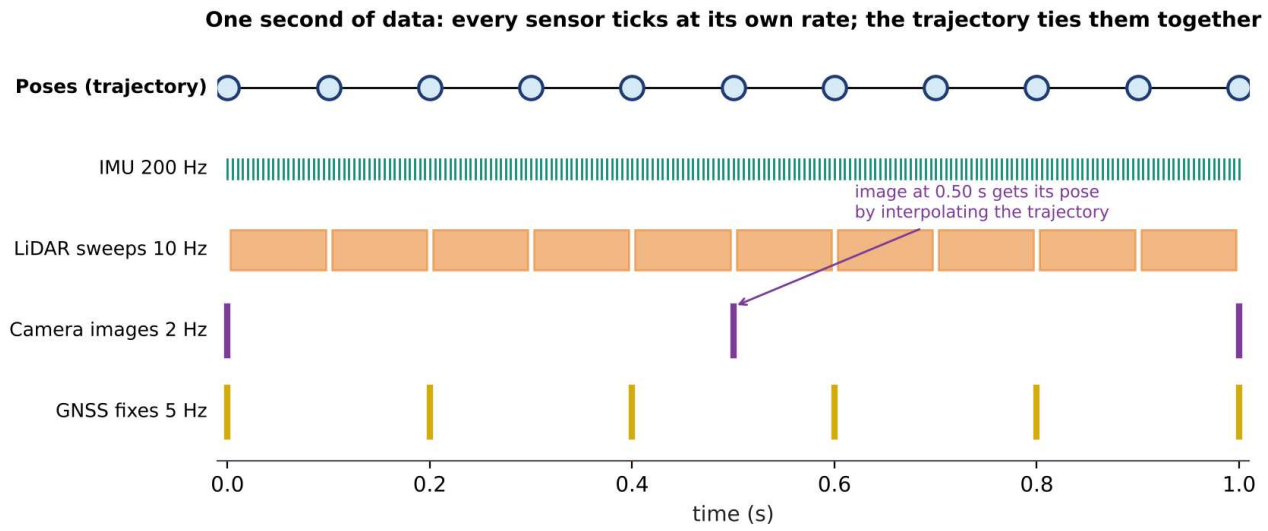
Book: Module 12 • Workbook: sheet M12 Factor Graphs

Hundreds of IMU samples become one factor



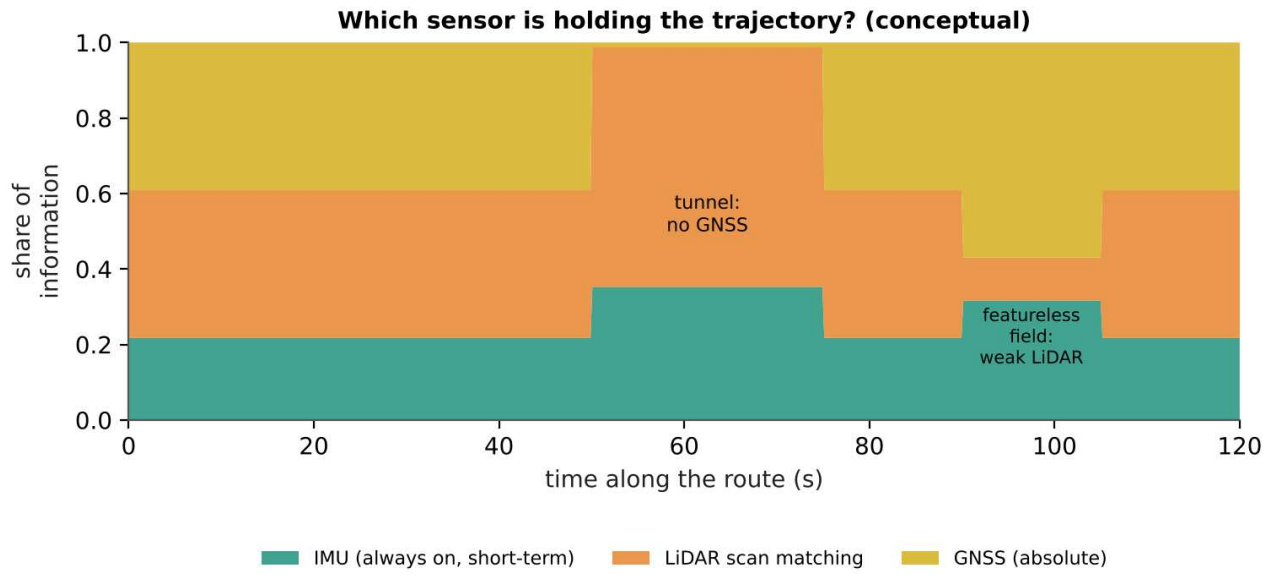
- Between two keyframes, the IMU samples are summarized once into a relative motion
- Worked in one dimension: $\Delta v = 0.050$ m/s, $\Delta p = 2.5$ mm
- The optimizer can move the poses without re-integrating the raw data

One second of data, one common thread



- IMU: 200 ticks. LiDAR: 10 sweeps. GNSS: 5 fixes. Camera: 2 images
- State: $\mathbf{x} = [\mathbf{R},\ \mathbf{p},\ \mathbf{v},\ \mathbf{b}]$
- The trajectory gives every measurement its pose
- Only keyframes enter the graph

The graph leans on whichever sensor has information



- Open sky: GNSS and IMU
- Tunnel: LiDAR matching and IMU
- Featureless field: GNSS and IMU
- Inspect the trajectory’s standard deviation before trusting any part of the cloud

WHAT THE ESTIMATOR DOES

0.186 m

from a 0.50 m prediction and a 0.20 m measurement: fusion beats the best source

0.20 m

of misclosure shared by four measurements: an adjustment

SLAM is a weighted least-squares adjustment over time. Its accuracy is limited by the largest unmodeled error, and a wrong weight is an unmodeled error.

SESSION 4 • 3:50–4:50 • MODULES 14

Error propagation and error budgets

The variance law, correlated errors, the covariance form, the full georeferencing budget, Monte Carlo checking, and the bias floor.

MODULE 14 • KEY EQUATION

The law of propagation of variances

$$\sigma_y^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2$$

Given	scanner 2.00 m above the road, point 20.00 m away; $z = \rho \sin V$
Inputs	$\sigma_\rho = 0.35$ cm, $\sigma_V = 0.01°$
Range part	$0.0995 \times 0.35 = 0.035$ cm
Angle part	$20.000 \times 1.745 \times 10^{-4} = 0.349$ cm
Combine	$\sqrt{0.035^2 + 0.349^2} = 0.351$ cm

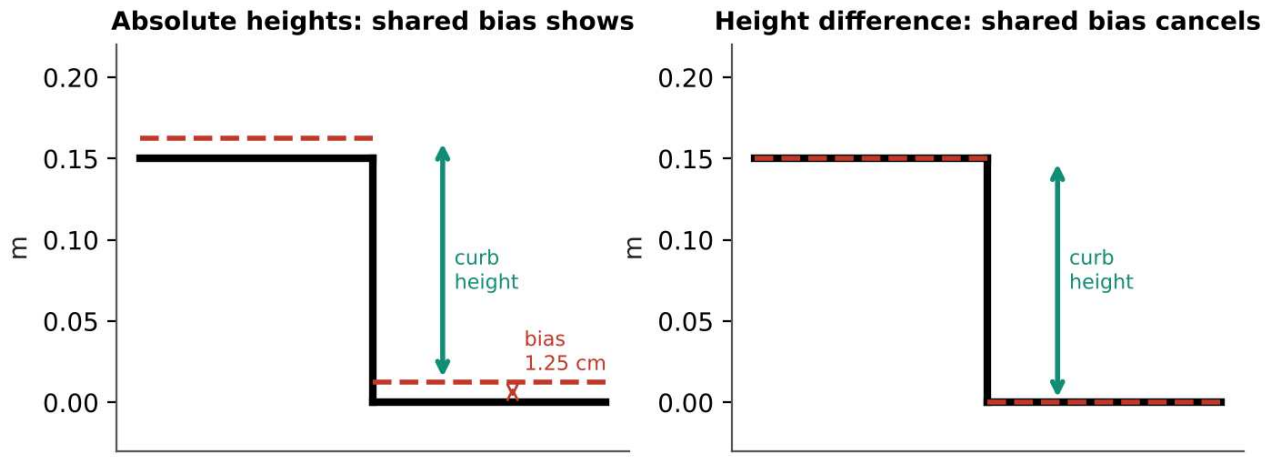
3.5 mm

height error of a road point at 20 m

- Each input’s spread, times how strongly the output responds to it
- On a grazing road surface the angle controls the height, not the range

Book: Module 14 • Workbook: sheet M14 Error Budgets

A shared bias cancels in a difference



3.5 mm

curb height, though each absolute height is good only to about 1.3 cm

- $\sigma_y^2 = \sum a_i^2 \sigma_i^2 + 2 \sum a_i a_j \, \sigma_{ij}$
- Same sweep, same 1.25 cm range bias on both points
- Relative measurements beat absolute ones

MODULE 14 • KEY EQUATION

Many outputs at once: the covariance matrix

$$\boldsymbol{\Sigma}_y = \mathbf{J}\,\boldsymbol{\Sigma}_x\,\mathbf{J}^{\mathsf{T}}, \qquad J_{ki} = \frac{\partial y_k}{\partial x_i}$$

Inputs	$\rho,\ H,\ V$ with their variances on the diagonal of $\boldsymbol{\Sigma}_x$
Jacobian	how each of X, Y, Z responds to each input
Output	a 3 by 3 covariance: an error ellipsoid
95% region	1-D: $1.960\,\sigma$ 2-D: $2.448\times$ semi-axes 3-D: $2.796\times$

2.448

turns a 1σ horizontal ellipse into a 95% ellipse

- The same law, written for three outputs
- When range noise dominates, the ellipsoid is a cigar pointing back at the scanner

Book: Module 14 • Workbook: sheet M14 Error Budgets

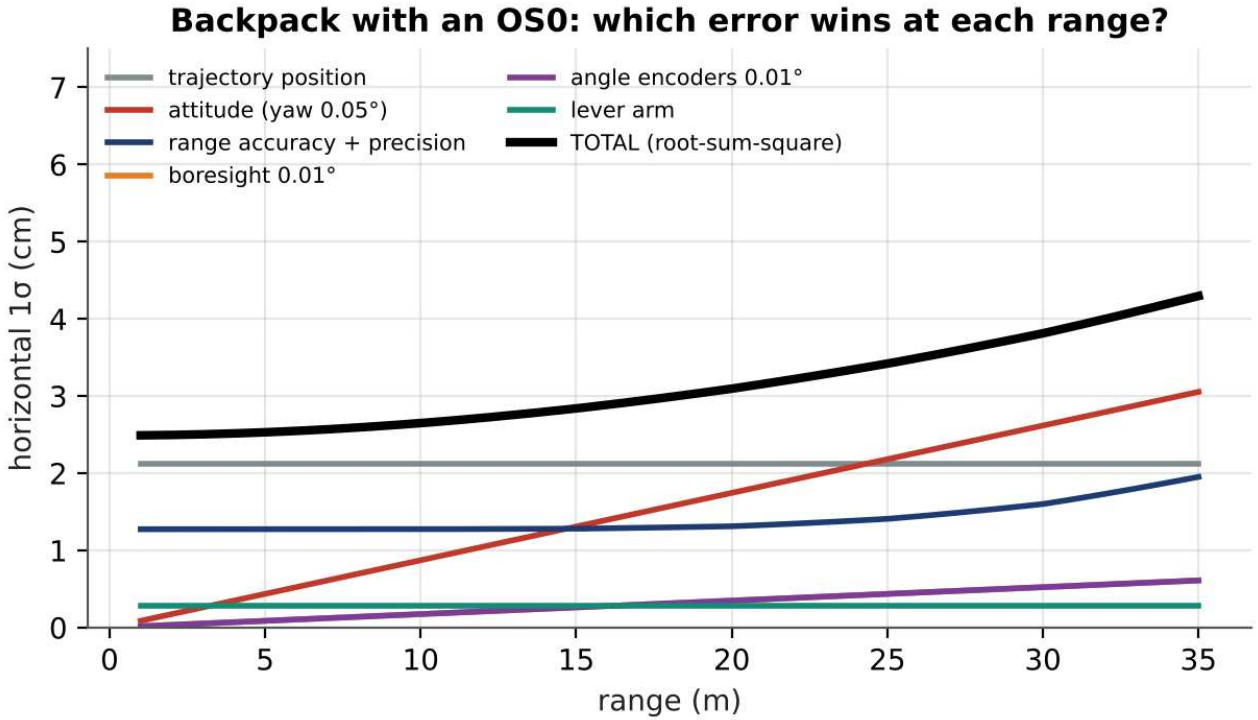
MODULE 14 • WORKED EXAMPLE

Full error budget of a backpack point at 20 m (1σ, cm)

Term	Horizontal	Vertical
Trajectory position (SLAM + control)	2.12	2.00
Attitude: yaw 0.05°, roll and pitch 0.02°	1.75	0.70
Boresight 0.01° per axis	0.35	0.35
Lever arm 2 mm per axis	0.28	0.20
Range: accuracy 1.25, precision 0.40	1.31	—
Angle encoders 0.01°	0.35	0.35
Total (root-sum-square)	**3.10**	**2.18**

Book: Module 14 • Workbook: sheet M14 Error Budgets

Fixed terms dominate close in; angle terms take over far out



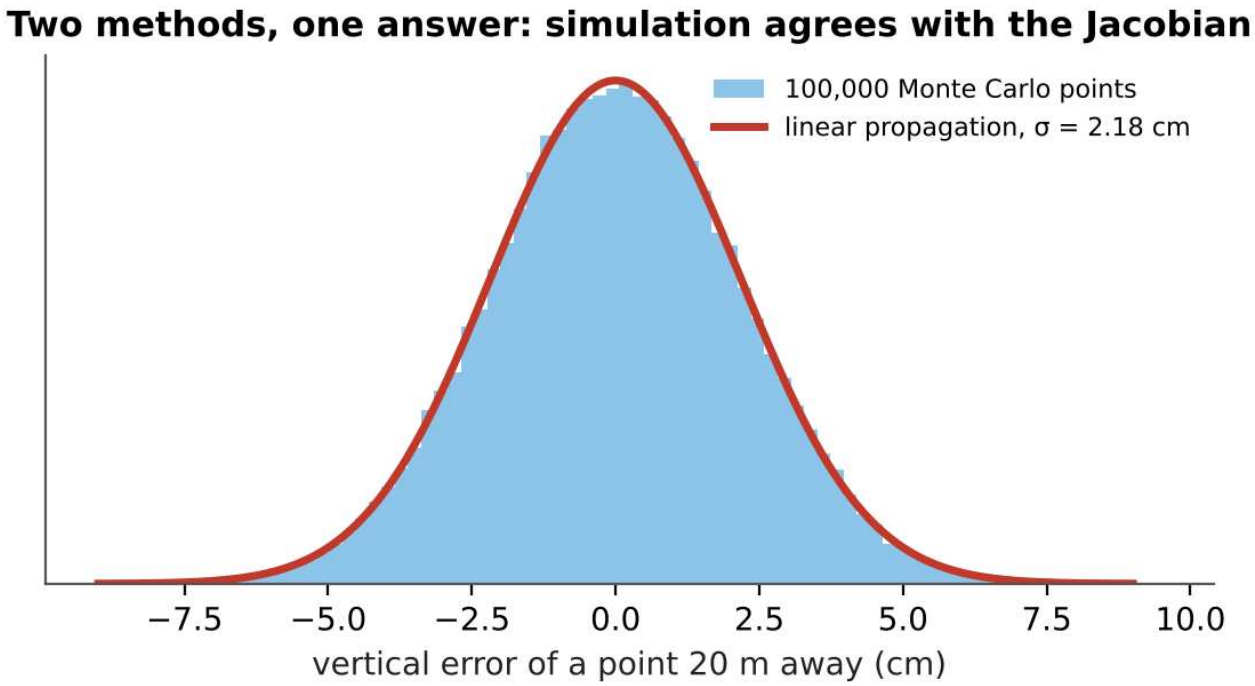
18 m

where this system stops meeting a 3 cm horizontal specification

- Use the curve to set the maximum useful range
- Beyond it: filter points by range, or improve heading

Book: Module 14 • Workbook: sheet M14 Error Budgets

Monte Carlo: simulate it 100,000 times

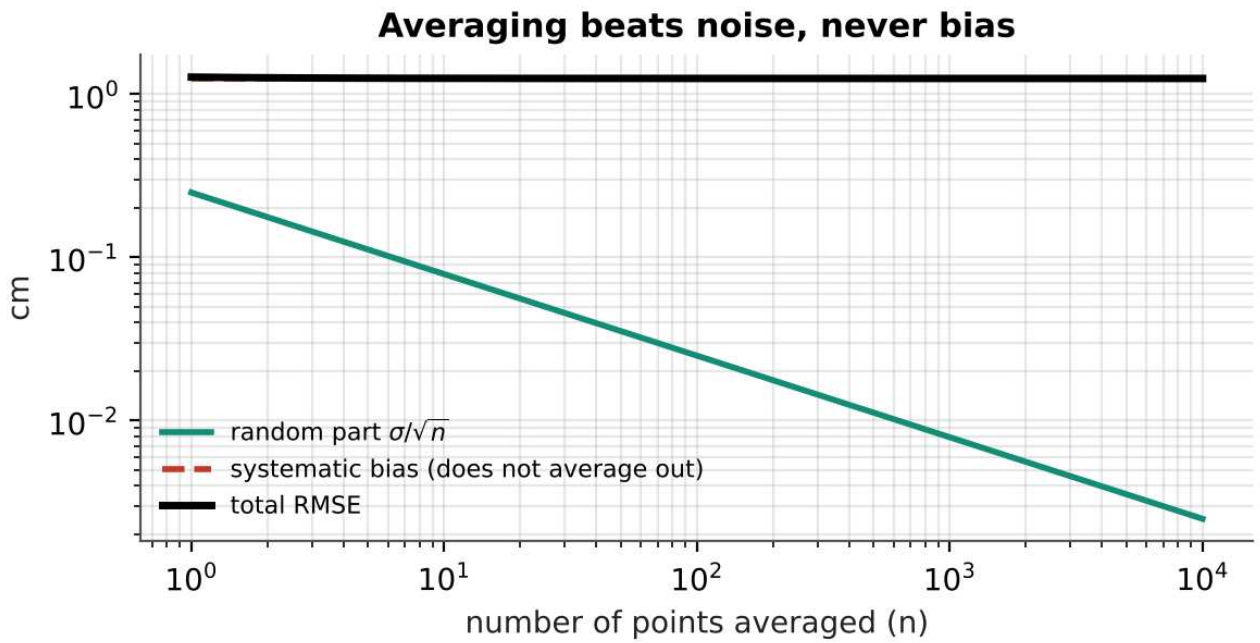


under 1%

difference between simulation and the linear formula

- Horizontal: 3.09 cm simulated, 3.10 cm by formula
- Vertical: 2.18 cm both ways
- No derivatives needed; handles any distribution

Averaging shrinks noise but not bias



1.25 cm

the floor: 400 returns on a target average the noise away, not the bias

- $\text{RMSE}_{\bar{x}} = \sqrt{b^2 + \sigma^2/n}$
- Random part: $0.25/\sqrt{400} = 0.0125$ cm
- Only calibration against known distances removes a bias

MODULE 14 • WORKED EXAMPLE

A root-sum-square budget for a 5 cm job (1σ, cm)

Term	As planned	Featureless stretch
Range and angle	2.0	2.0
Lever arm and boresight	1.5	1.5
Trajectory	3.0	**5.0**
Timing	0.5	0.5
Control and georeferencing	1.5	1.5
Total	**4.21: passes**	**5.81: fails**

THE BUDGET IN TWO NUMBERS

3.10 cm

horizontal, 1σ: backpack point at 20 m

2.18 cm

vertical, 1σ: the same point

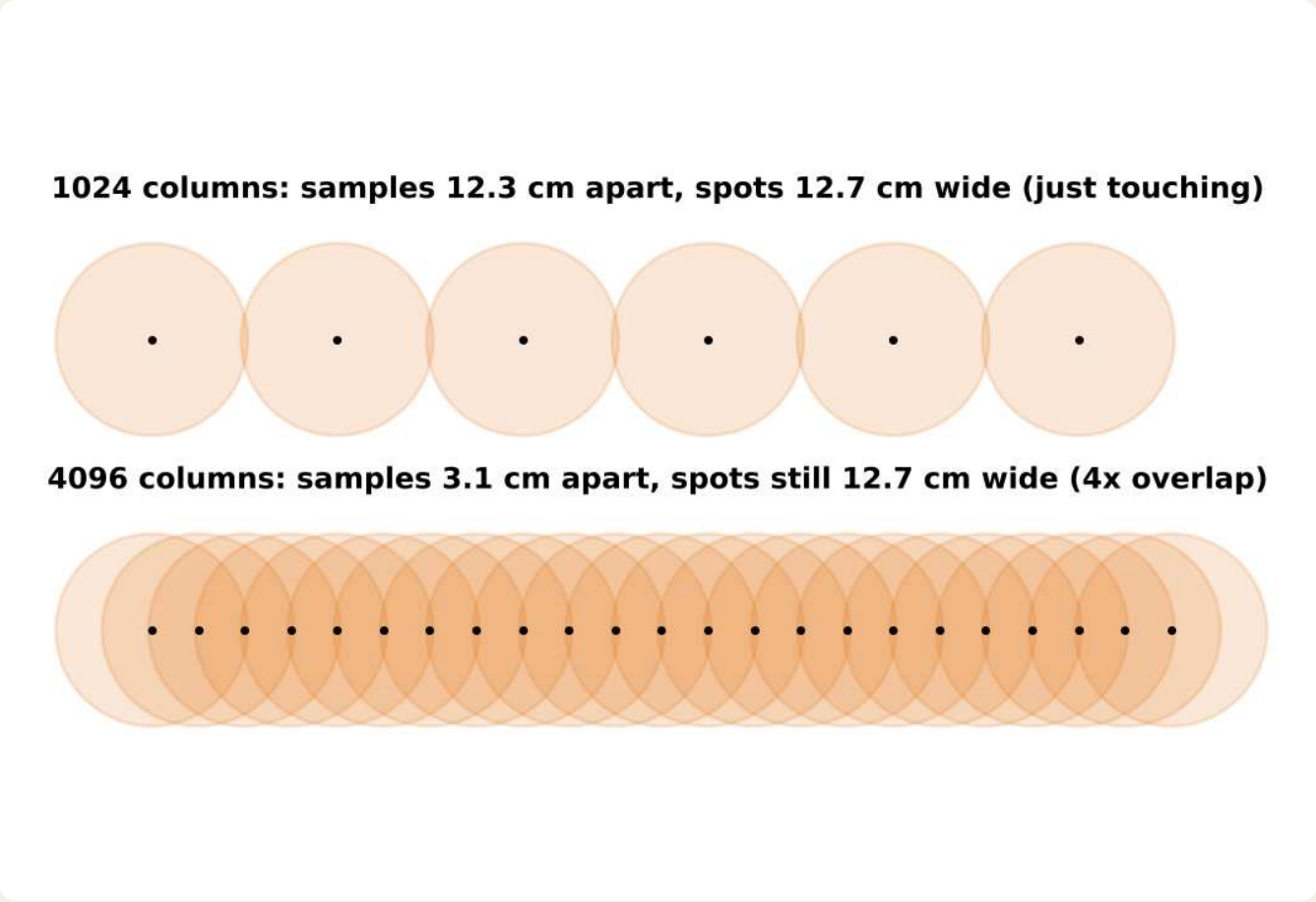
Independent errors combine by root-sum-square, so the largest term dominates. Find it, and spend the effort there.

SESSION 5 • 4:50–5:35 • MODULES 15, 16, 17

Hardware in numbers

Reading a scanner datasheet, choosing an IMU by its coast error, and a survey-grade airborne system at 500 ft.

More points is not the same as sharper detail

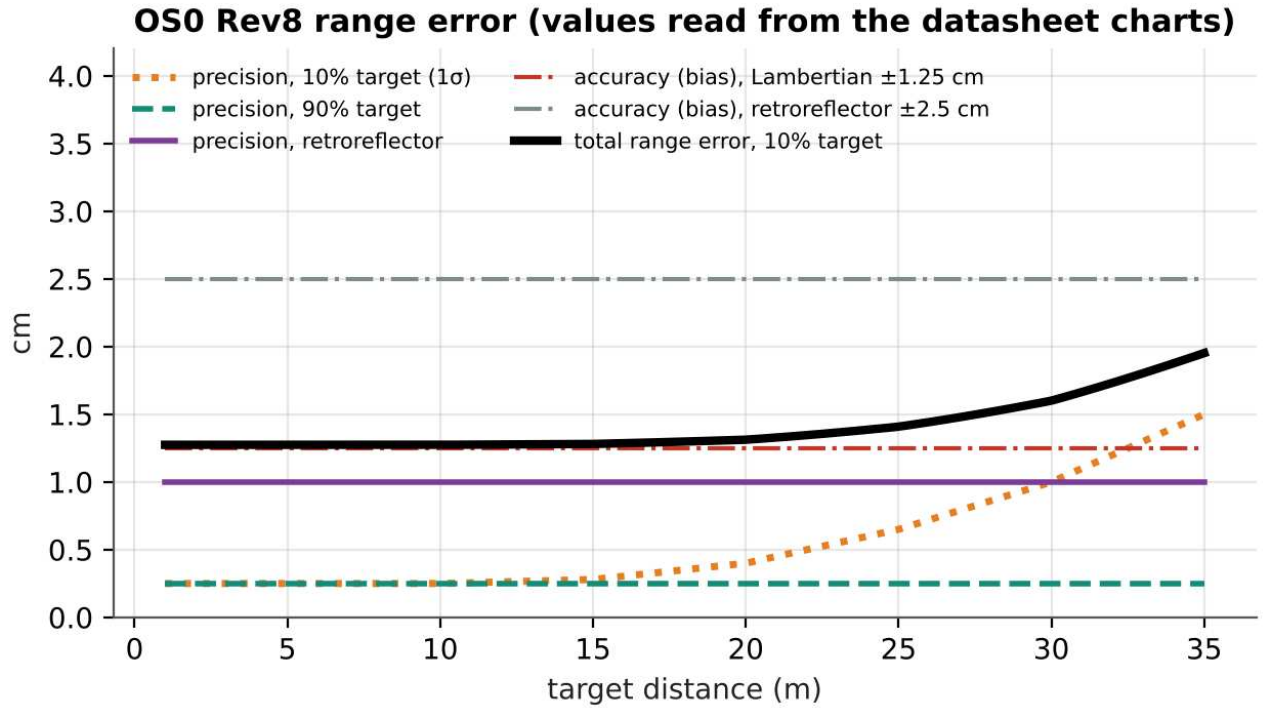


12.7 cm

spot size at 20 m for a 0.35° beam

- 1024 mode at 20 m: 12.3 cm by 24.7 cm grid in one sweep
- 4096 mode: 3.1 cm spacing, four samples inside every spot
- Edges are located only to about half a spot: 6 cm

Total range error from a datasheet



1.25 cm

range bias: it dominates out to about 25 m

- $\sigma_\rho \approx \sqrt{b^2 + \sigma_p^2 + \sigma_q^2}$
- Dark asphalt: 1.27 cm at 10 m, 1.31 at 20 m, 1.60 at 30 m, 1.95 at 35 m
- Retroreflective targets are worse: use matte targets for height

MODULE 16 • KEY EQUATION

Coast error through a GNSS outage

$$\delta x(t) \approx \sqrt{(\sigma_v t)^2 + \left(\tfrac{1}{2} b_a t^2\right)^2 + \left(\tfrac{1}{6} g\, b_g t^3\right)^2}$$

Low-cost, 20 s	2.07 m: low-cost GNSS/INS
RTK INS, 20 s	1.00 m with datasheet velocity
RTK velocity	0.22 m after 20 s
60 s outage	15.0 m against 2.64 m

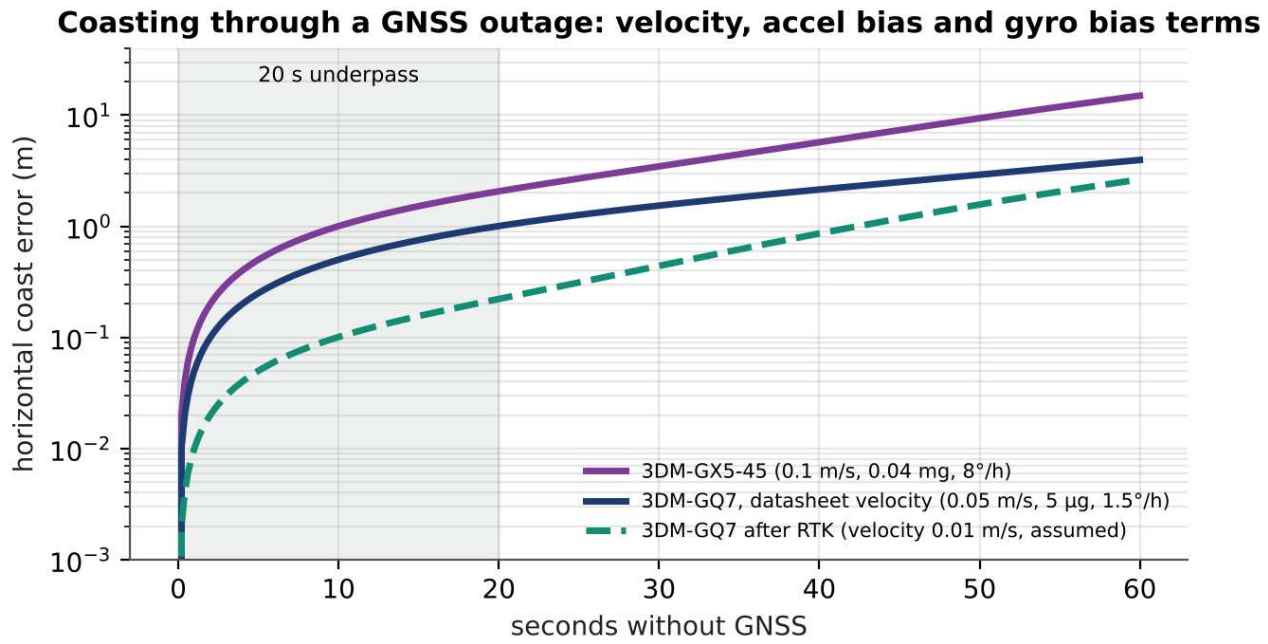
0.22 m

best case after 20 s without GNSS, MEMS-grade INS

- Velocity error first, then accelerometer bias, then the gyro’s cubic term
- No MEMS INS holds centimeters through a minute-long tunnel

Book: Module 16 • Workbook: sheet M16 GX5-GQ7

How long an outage can this IMU carry?



- Read the curve at your tolerance, not at the brochure number
- A 300 m underpass at 13 m/s is a 23 s outage
- Scan matching and control inside the outage do the rest
- State the IMU grade in the scope

MODULE 16 • KEY EQUATION

Dual-antenna heading improves with a longer baseline

$$\sigma_\psi \approx \frac{\sigma_{\text{rel}}}{b}$$

Datasheet	0.25° at 1.5 m gives $\sigma_{\text{rel}} = 6.5$ mm
Backpack 0.5 m	$0.0065/0.5 = 0.75°$
Vehicle 2.5 m	$0.0065/2.5 = 0.15°$

0.15°

on a 2.5 m vehicle baseline, against 0.75° on a backpack

- Immune to steel, and works at a standstill
- Still 13 cm at 50 m: scan matching must tighten it

Where the error comes from (1σ, cm, nadir return)

Term	Horizontal	Vertical
GNSS position	2.5	3.75
Roll and pitch 0.005°	1.88	about 0
Boresight 0.005° per axis	1.88	—
Laser range	—	2.5
Lever arm 5 mm	0.71	0.5
Total	**3.7 (0.12 ft)**	**4.5 (0.15 ft)**
With control and strip adjustment	**3.1**	**3.1**

WHAT THE HARDWARE NUMBERS SAY

GNSS

is the largest term for a survey-grade system at 500 ft

4.6 cm

atmospheric correction at 500 ft: confirm it is switched on

62.4 m

ambiguity zone at 2.4 MHz: a wrong zone is a blunder

Survey-grade results come from the whole system: trajectory, calibration, control and checkpoints. Not from the laser alone.

SESSION 6 • 5:45–6:35 • MODULES 13, 19, 21

The standards in numbers

ASPRS Edition 2 and FDOT Type A, B and C: converting between them, running one test for both, and designing the control to pass.

MODULES 13 AND 19 • TWO STANDARDS

ASPRS Edition 2 and FDOT test accuracy differently

	ASPRS Edition 2 (2024)	FDOT Handbook (2025), TML
Statistic	RMSE, with the mean error stated	95% confidence (NSSDA)
Checkpoints	at least 30, up to 120	at least 20
Checkpoint accuracy	at least 2× better than the product	the PSM's choice (Note 5)
Checkpoint error	included in the result	not required
Relative accuracy	numeric limits within and between swaths	20% overlap
Vegetation	VVA reported as found	not addressed

Book: Modules 13 and 19 • Supplementary: FDOT Handbook Sections 39.6.2 and 39.7

What each FDOT type needs as an RMSE

FDOT type	95% tolerance, H and V	$RMSE_V$ needed ($\div 1.96$)	$RMSE_H$ needed ($\div 1.7308$)
A	0.06 ft (1.83 cm)	0.031 ft (0.93 cm)	0.035 ft (1.06 cm)
B	0.10 ft (3.05 cm)	0.051 ft (1.56 cm)	0.058 ft (1.76 cm)
C	set by the PSM and the scope	—	—

MODULE 13 • LAB 5

The checkpoint test, both ways

$$\text{RMSE} = \sqrt{\frac{\sum e_i^2}{n}}, \qquad \text{RMSE}_{\text{product}} = \sqrt{\text{RMSE}_{\text{fit}}^2 + \text{RMSE}_{\text{survey}}^2}$$

Ten residuals	mean $+0.53$ cm: no meaningful bias
RMSE	$\sqrt{65.73/10} = 2.56$ cm
Plus survey error	$\sqrt{2.56^2 + 1.0^2} = 2.75$ cm
FDOT 95%	$1.96 \times 2.75 = 5.39$ cm $= 0.177$ ft
Verdict	fails Type A (0.06 ft) and Type B (0.10 ft)

2.75 cm

RMSE_V with checkpoint error: 0.177 ft at 95%

- Report the mean as well as the RMSE
- Ten points is a field check; a formal test needs 30

Book: Module 13 and Lab 5 • Workbook: sheet M13 Accuracy

Passes its ASPRS agreement. Meets neither FDOT type.

Item	Sample project	Type A	Type B
Vertical at 95%	$1.96 \times 2.06 = 4.04$ cm (0.132 ft)	≤ 0.06 ft: **fails**	≤ 0.10 ft: **fails**
Horizontal at 95%	$1.7308 \times 2.82 = 4.88$ cm (0.160 ft)	≤ 0.06 ft: **fails**	≤ 0.10 ft: **fails**
Point density	58 points/ft^2	≥ 20: meets	≥ 10: meets
Checkpoints	30, about 220 ft apart	meets	meets
GNSS	300 m outage, about 85% fixed	document (Note 4)	document (Note 4)

ASPRS data internal precision limits (cm)

Vertical accuracy class	Within swath	Between swaths, RMSD	Between swaths, max
Factor of the class X	$0.60X$	$0.80X$	$1.60X$
1 cm (near Type A)	0.6	0.8	1.6
1.5 cm (near Type B)	0.9	1.2	2.4
3 cm (the capstone)	1.8	2.4	4.8
5 cm (a Type C choice)	3.0	4.0	8.0

Book: Module 13, ASPRS Edition 2 Version 2, Table 7.2

RECOMMENDATIONS DOCUMENT, PROBLEM 5

Control spacing for Type A: by conditions, not one number

Conditions	Maximum control spacing
FDOT recommendation, any conditions	1000 ft
Open sky, clean fixed solution throughout	500 ft
Tree canopy, buildings, intermittent fix	300 ft
GNSS outage (overpass, tunnel)	150–300 ft, with at least one point inside

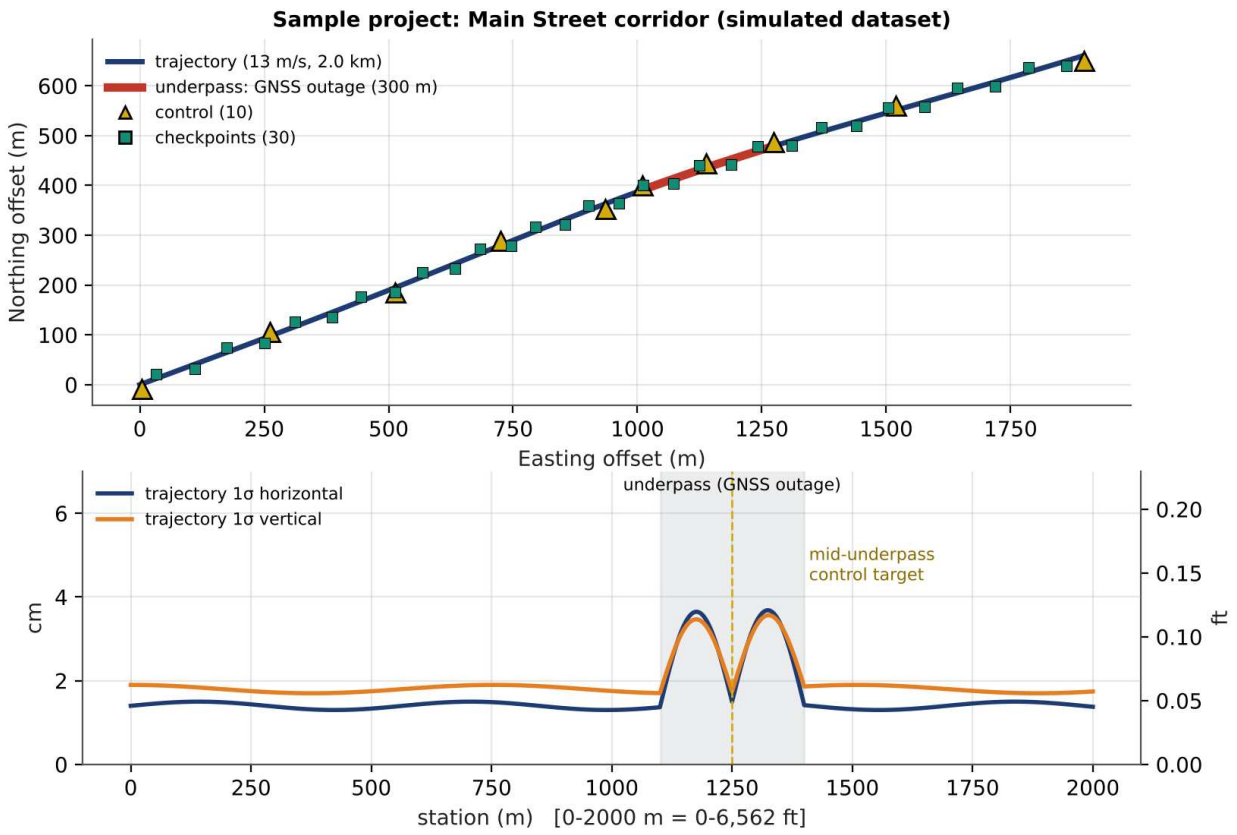
RECOMMENDATIONS DOCUMENT, PROBLEM 5

Scans between control points behave like traverse legs

Speed	300 ft spacing	1000 ft spacing	Peak at the midpoint
13 m/s (29 mph)	about 70 scans, 7 s	about 235 scans, 23 s	about 4 mm against 8 mm
5 m/s (11 mph)	about 180 scans, 18 s	about 610 scans, 61 s	more scans, more drift time

Assumes 1 mm of independent relative error per scan, for illustration • 10 Hz scanner

Read the trajectory before the point cloud



2.23 cm

vertical budget in the underpass with a control target inside it

- Open sky: 2.45 cm. Underpass: 3.00 cm, at the limit
- Place control where the trajectory's own uncertainty rises
- Put checkpoints midway, and inside every outage
- Collect each stretch in both directions

Book: Modules 14 and 21 • Workbook: sheets M21 Capstone and Dataset

ONE TEST FOR BOTH STANDARDS

A checkpoint test that satisfies ASPRS and FDOT

- At least 30 checkpoints, independent of all control
- Surveyed at least 2× better than the tolerance: about 5 mm for Type A
- Placed midway between control points and inside every GNSS outage
- Horizontal tested on fitted targets sized for the point spacing
- Report the mean error, the RMSE and the 95% value
- Include the checkpoint survey error in the result
- Test swath-to-swath agreement against the ASPRS limits
- State datum, epoch, geoid, foot, and the fixed-solution percentage

Supplementary: accuracy testing SOP • Recommendations document, summary table

THE THREE NUMBERS TO TAKE HOME

0.93 cm

$RMSE_V$ that FDOT Type A needs

30

checkpoints: one test that satisfies ASPRS and FDOT

300 ft

default control spacing for Type A where GNSS is weak

Type A is a control-driven survey. Budget the uncertainty, put control where the trajectory is weak, and prove it with independent checkpoints.

6:35–7:05

Problem set, and what comes next

- 20 worked problems, open book
- Calculator and the workbook allowed
- At least two problems from each session
- Each asks for a number and a one-line conclusion
- 70% to pass: 14 of 20

- Seminar 3: five projects from scope to signed deliverable
- Type A resurfacing, Type B inventory, an outage, Type C, an acceptance test
- Bring your own project data if you can share it